

Hoang Anh (Ben) NGUYEN

Email: hoanganh.nguyen@mines.edu | Personal Website: hnguyenx.com | [Google Scholar](#)

RESEARCH SUMMARY

I am a PhD student in Geophysics at the Colorado School of Mines. My research focuses on quantum computing algorithms for solving PDEs and inverse problems. I also work on large-scale seismic multi-component elastic full waveform inversion (FWI) with GPU-accelerated solvers, as well as scientific computing and machine learning applications for geophysics.

EDUCATION

PhD in Geophysics | Major: Geophysics, Minor: Computer Science Aug 2023 – May 2027 (expt.)
Colorado School of Mines (CSM) Golden, CO, USA

Thesis: Solving inverse problems with quantum computing

Postgraduate Diploma in Earth System Physics | Major: Solid Earth Sept 2022 – Aug 2023
International Centre for Theoretical Physics (ICTP) - UNESCO Trieste, Italy

Thesis: Ambient noise tomography beneath the Banda Arc

Master of Engineering Physics | Major: Computational Physics May 2021 – Sept 2023
Hanoi University of Science and Technology (HUST) Hanoi, Vietnam

Thesis: Structural simulation of MgSiO_3 under compression

Erasmus Exchange | Major: Physics of Complex Systems Feb 2022 – Jul 2022
Polytechnic University of Turin (POLITO) Torino, Italy

Bachelor of Engineering Physics | Major: Computational Physics Sept 2016 – April 2021
Talent Honours Program – HUST Hanoi, Vietnam

Thesis: Computational modeling of microstructure of magnesium silicate

RESEARCH EXPERIENCE

Research Assistant Aug 2023 – Present
Department of Geophysics, CSM Advisor: Prof. Tura, A.

Project: End-to-end quantum algorithms for hyperbolic PDEs

- Develop explicit-circuit quantum algorithms for acoustic and elastic wave propagation based on Hamiltonian simulation and Schrödingerisation, with convolutional perfectly matched layer (C-PML) absorbing boundaries.
- Extend the formulation to the elastic velocity-stress system on staggered grids. Reflection levels reach order 10^{-4} , on par with classical C-PML implementations.

Project: Gate-based quantum computing for the forward modeling and inverse problems

- Design a hybrid quantum-classical finite-basis physics-informed neural networks (PINNs) for FWI, in which the domain-decomposed wavefield networks and the global velocity network terminate in parameterized quantum circuits.
- Implement the circuits as differentiable JAX statevector simulators. Automatic differentiation then runs end to end through the network, the circuit, and the physics-informed loss, so no parameter-shift rules are needed.
- Reach lower velocity error than the classical baseline in roughly $8\times$ fewer training iterations, with about 33% fewer trainable parameters. The hybrid model outperforms all other classical hyperparameter variants tested.

Project: Optimization with quantum annealing

- Reformulate regularized least-squares problems as quadratic unconstrained binary optimization (QUBO) and solve them on the D-Wave Advantage annealer. Large systems are split into independent sub-problems that fit the hardware qubit limits.
- Benchmark solution accuracy and sensitivity to data noise against classical Tikhonov-regularized least squares.

Project: GPU-accelerated multi-component elastic FWI

- Lead the FWI team across code development, research directions, and results delivery.

- Build **xFWI**, a GPU-accelerated multi-component elastic FWI framework in Devito for multi-component geophone and distributed acoustic sensing data, with perfectly matched layer absorbing boundaries. The GPU implementation runs up to 100× faster than a baseline multi-core CPU version.
- Implement wavefield checkpointing for the adjoint-state gradient. Memory use stays bounded, so models larger than GPU memory can still be inverted.

Project: Time-lapse FWI for CO₂ sequestration monitoring

- Apply time-lapse FWI to baseline and monitor distributed acoustic sensing surveys to image CO₂ plume evolution and quantify time-lapse velocity and density changes.

Project: Spectral element modeling and FWI with automatic differentiation

- Develop **torchchem**, a PyTorch spectral-element wave propagation framework for acoustic, elastic, SH, and coupled fluid–solid media, with anisotropy, constant-*Q* attenuation, and C-PML absorbing boundaries. All kernels run on CPU and GPU, and are validated against analytical solutions and **SPECFEM** to cross-correlation above 0.9999.
- Keep the solver end-to-end differentiable so that FWI gradients follow from automatic differentiation through the time-stepping loop, without hand-derived adjoint kernels.

Research Assistant

May 2023 – Aug 2023

Earth System Physics, ICTP

Advisors: Prof. Aoudia, A. & Dr. Manu-Marfo, D.

Project: Ambient noise tomography beneath the Banda arc–continent collision zone

- Developed MPI-parallelized ambient noise tomography workflows on four years of continuous seismic data. Extracted empirical Green's functions, inverted surface-wave dispersion for shear-wave velocity down to 200 km depth, and assessed model resolution.

Research Assistant

Aug 2018 – July 2023

Department of Computational Physics, HUST

Advisor: Prof. Nguyen, V.H.

Project: Molecular dynamics of silicate, germanate, and phosphate glasses under compression

- Developed parallel molecular dynamics codes in C on RIKEN's **Fugaku** supercomputer to simulate silicate, germanate, and phosphate glasses and liquids under compression. Characterized pressure-induced coordination and topological transitions.

Project: First-principles modeling of hydrogen storage alloys

- Performed first-principles density functional theory calculations with **Quantum ESPRESSO**. Evaluated electronic structure and formation enthalpies using plane-wave pseudopotential methods.

WORK EXPERIENCE

Geophysics R&D Research Intern

May 2026 – Aug 2026

Shell: global integrated energy company

Houston, TX

Project: Wave propagation and FWI across undulating interfaces

- Reproduced and extended the fractional grid method so that interfaces lying between grid nodes sit at their true depth in finite-difference modeling.
- Built spectral-element solutions on interface-conforming curved meshes.
- Ran acoustic and elastic FWI on a high-performance computing (HPC) cluster to test how the sub-cell model-building recipe changes the inversion, from data generation on a fine reference model through inversion of coarse-grid start models.

Geophysics R&D Research Intern

May 2025 – Aug 2025

TGS: energy data and analytics company

Houston, TX

*Project: GPU-accelerated *Q*-attenuation modeling and imaging in anisotropic media*

- Developed GPU-accelerated seismic wave solvers for high-performance computing environments.
- Modeled *Q*-attenuation and compensation in isotropic and anisotropic (VTI/TTI) media using Devito Pro.

- Formulated an anisotropic attenuation model on fully staggered grids.
- Implemented Q-elastic gradient computation for reverse time migration (RTM) and FWI.

Geophysics R&D Research Intern

VPI: national energy research institute

Sept 2021 – Feb 2022

Hanoi, Vietnam

Project: 3D ray tracing and traveltimes tomography

- Implemented 3D seismic ray-tracing algorithms for traveltimes modeling in heterogeneous velocity models.
- Applied non-linear traveltimes tomography techniques for subsurface velocity reconstruction.

TEACHING

Teaching Assistant | GPGN 536: Advanced Geophysical Computing

Fall 2026

Department of Geophysics, CSM

Golden, CO

- Deliver selected lectures and lead hands-on sessions on computational methods for geophysical problems.

CURRENT WORK & PUBLICATIONS

- [1] Nguyen, H.A., Tura, A. Convolution absorbing boundaries for explicit-circuit quantum computing of the wave propagation. In preparation.
- [2] Nguyen, H.A., Vashisth, D., Tura, A. Accelerating physics-informed neural networks for full waveform inversion using a hybrid quantum-classical finite-basis architecture. Under review at *IEEE Transactions on Geoscience & Remote Sensing* (2026). arxiv.org/abs/2606.01110
- [3] Nguyen, H.A., Tura, A. Joint multi-parameter elastic full-waveform inversion using distributed acoustic sensing and multi-component seismic data. Submitted to *Computers & Geosciences* (2026). arxiv.org/abs/2607.01649
- [4] Nguyen, H.A., Manu-Marfo, D., & Aoudia, A. Ambient noise tomography beneath the Banda basin reveals new insights into the arc-continent collision zone. Under review at *Geophys. Res. Lett.* (2026). [10.22541/es-soar.176790173.31128656/v1](https://doi.org/10.22541/es-soar.176790173.31128656/v1)
- [5] Plan, E.L.C.V.M., Phan, H., Nguyen, H.A. *et al.* Numerical simulation on structural and topological transitions of GeO₂ liquid under compression. *Eur. Phys. J. B* **99**, 20 (2026). [10.1140/epjb/s10051-026-01143-0](https://doi.org/10.1140/epjb/s10051-026-01143-0)
- [6] Nguyen, H.A., Tura, A. Seismic traveltimes inversion with quantum annealing. *Scientific Reports* **15**, 17984 (2025). [10.1038/s41598-025-01188-8](https://doi.org/10.1038/s41598-025-01188-8)
- [7] Lai, D.V., Nguyen, H.S., Nguyen, H.A. *et al.* Tailoring hydrogen storage performance of Mg–Mg₂Ni alloys: synergistic effects of composition and phase formation with first-principles insights. *RSC Advances* **15**, 31240–31254 (2025). [10.1039/D5RA04356E](https://doi.org/10.1039/D5RA04356E)
- [8] Nguyen, V.H., Nguyen, H.A. Crystallisation of liquid silica under compression: a molecular dynamics simulation. *Pramana - J Phys* **98**, 142 (2024). [10.1007/s12043-024-02839-7](https://doi.org/10.1007/s12043-024-02839-7)
- [9] Nguyen, H.A., Nguyen, V.H. Study of the structure of MgSiO₃ system under compression by using ring statistics and Voronoi analysis. *Phys. Scr.* **98**, 045919 (2023). [10.1088/1402-4896/acc5b7](https://doi.org/10.1088/1402-4896/acc5b7)
- [10] Nguyen, V.H., Nguyen, H.A., Iitaka, T., Mai, T.L. Computer simulation of phosphate-silicate and calcium phosphate-silicate systems. *Phys. Scr.* **98**, 065704 (2023). [10.1088/1402-4896/acd4fb](https://doi.org/10.1088/1402-4896/acd4fb)
- [11] Nguyen, H.A., Nguyen, H.S., Nguyen, V.H. Pressure-induced glassy networks of enstatite (MgSiO₃) and forsterite (Mg₂SiO₄). *VNU J. Sci. Math. - Phys.* **39**, 1 (2023). [10.25073/2588-1124/vnumap.4767](https://doi.org/10.25073/2588-1124/vnumap.4767)
- [12] Pham, T.H.H., Doan, H.H., Ta, Q.M., Mai, T.L., Nguyen, H.A. Some results of seismic travel-time reflection tomography study. *Petrovietnam Journal* **10**, 4–16 (2021). [10.47800/PVJ.2021.10-01](https://doi.org/10.47800/PVJ.2021.10-01)
- [13] Nguyen, H.S., Nguyen, H.A., Pham, H.K., Iitaka, T., Nguyen, V.H. Topology of SiO_x units and glassy network of magnesium silicate glass under densification: correlation between radial distribution function and bond angle distribution. *Modelling Simul. Mater. Sci. Eng.* **28**, 065007 (2020). [10.1088/1361-651X/ab9bb4](https://doi.org/10.1088/1361-651X/ab9bb4)
- [14] Nguyen, H.S., Nguyen, H.A. Structural simulation of Mg₂SiO₄ under compression. *VNU J. Sci. Math. - Phys.* **36**, 4 (2020). [10.25073/2588-1124/vnumap.4484](https://doi.org/10.25073/2588-1124/vnumap.4484)

INTERNATIONAL TALKS AND PRESENTATIONS

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| [1] IMAGE26 (oral): Nguyen, H.A., Tura, A. Full waveform inversion with hybrid quantum finite-basis physics-informed neural network | Aug 2026
Houston, TX |
| [2] EAGE26 (workshop): Tura, A., Yermakov, R., Nguyen, H.A., Sabbagh, R. Leveraging multicomponent OBN acquisitions for improved PP-PS imaging and inversion | Jun 2026
Aberdeen, Scotland |
| [3] IAGA-IASPEI25 (poster): Manu-Marfo, D., Nguyen, H.A., Aoudia, A. Ambient noise tomography beneath the Banda basin reveals new insights into the arc-continent collision zone | Sep 2025
Lisbon, Portugal |
| [4] IMAGE25 (workshop): Nguyen, H.A. Optimization with quantum annealing method | Aug 2025
Houston, TX |
| [5] IMAGE25 (oral): Nguyen, H.A., Tura, A. Crosswell traveltimes inversion using a quantum computing method | Aug 2025
Houston, TX |

SERVICE AND MENTORING

- [1] Peer-review service
- Journal of Computational Physics (Elsevier)
 - Geophysical Journal International (Oxford)
 - IMAGE (SEG)
 - Acta Geophysica (Springer)
- [2] PhD application mentoring: mentored 4 applicants, all of whom received PhD offers:
- Department of Material Design and Innovation, University at Buffalo
 - Department of Earth and Atmospheric Sciences, University of Houston
 - Department of Civil and Environmental Engineering, University of Southern California
 - Department of Earth and Planetary Sciences, Jackson School of Geosciences, University of Texas at Austin

HONORS AND AWARDS

- [1] Fully funded scholarship for postgraduate program at ICTP 2023
- [2] VEF 2.0 Program recommended candidate 2023: The [VEF 2.0](#) Program is conducted by the Fellows and Scholars of the Vietnam Education Foundation (VEF) – an independent U.S. Federal Government agency created by the U.S. Congress
- [3] Erasmus scholarship for master exchange students at POLITO 2022
- [4] Fully funded scholarship for the master program at HUST 2021, 2022
- [5] Certificate of Merit from School of Engineering Physics for undergraduates: Excellent Student in Fall Semester 2017, Spring Semester 2018, Fall Semester 2018, Spring Semester 2019
- [6] The 20th Vietnam National Student Physics Olympiad 2018: Second Prize
- [7] Lawrence S. Ting Scholarship 2017 for undergraduates
- [8] Vietnam Robot National Contest 2019: team member of HUST

CERTIFICATIONS

- [1] Deep Learning Specialization from DeepLearning.AI ([Certificate](#))
- [2] Quantum Computing and Quantum Information from IBM ([Certificate](#))

TECHNICAL SKILLS

Programming: Python, Julia, C, C++, Fortran, MATLAB
Machine Learning Frameworks: JAX, PyTorch, TensorFlow, scikit-learn
Geophysical Modeling: Anisotropic elastic FWI, RTM, DAS FWI, ambient noise tomography, quantum PDE solvers
Optimization: Adjoint-state methods, quantum annealing
HPC: Slurm, MPI, OpenMP/OpenACC, GPU computing